

Production-line assignment — Order impact design

005_DD-ORD version 1 is the WI-009-owned addendum for SCR-001 (order create/edit) and SCR-002 (order list), implementing FN-036 and REQ-068/069 with history rules from REQ-067. Approved plan revision 1 step 10. It extends the existing screens; it introduces no new route or independent DD family.

1. Document control and authoritative inputs

Field	Value
Document ID / version	005_DD-ORD / 1
System / subsystem	ProductionManagementAI / Production orders
Work item / author	WI-009 / Agent
Created / updated	2026-10-01 / 2026-10-01
Review state	Submitted for review; design only, not implemented

Version	Date	Author	Change
1	2026-10-01	Agent	Initial order picker/history/list impact and visuals

Inputs: [005_BD](#), [005_DB](#), [main DD](#), [API version 2](#), [FN version 1](#), [SPD version 1](#), [requirements](#), [decisions](#) and [plan](#).

Read-only baselines: [001_DD](#), [002_DD](#) and [WI-006 impact addendum](#). Original quantity, product retirement, status, due-date, notes, numeric order version, navigation and save-success behavior remain authoritative. Only line-related additions are owned here. Endpoint contracts remain in 005_DD-API section 6; backend validation/locks remain in 005_DD-FN section 6. All four 005 DD family members already exist; this addendum does not duplicate them or rewrite completed designs.

2. Screen layout and static review artifacts

Mockup artifact: [English gallery](#) and [Japanese edition](#). Native design/Artifact publishing remains unavailable; local HTML follows the approved design workflow fallback. The gallery has illustrative static states, Japanese UI and translated captions; it calls no APIs and records no runtime evidence. Reuse gray/white surfaces, gray-900 primary actions, Japanese fonts and responsive composition from ProductionOrderForm/Table.

1 ProductionManagementAI 製造指示一覧 生産ライン・工程

製造指示 PO-2026-0001

2 製品：P-1001 ブレーキキャリパー

数量：100 個 ステータス：下書き

納期：2026/10/05

3 生産ライン：L-001 組立ライン A

下書きでは任意です。

生産開始時に必要です。 選択を解除

4 ラインコード・ライン名で検索

L-003 加工ライン C 2.5 分／個

選択 前へ 1 / 1 次へ

5 エラー／保存結果

保存 キャンセル

1 PMAI メニュー

製造指示 PO-2026-0001

2 製品：P-1001 ブレーキキャリパー

数量：100 個 ステータス：下書き

納期：2026/10/05

3 生産ライン：L-001 組立ライン A

下書きでは任意です。

生産開始時に必要です。 選択を解除

4 ラインコード・ライン名で検索

L-003 加工ライン C 2.5 分／個

選択 前へ 1 / 1 次へ

5 エラー／保存結果

保存 キャンセル

1 ProductionManagementAI 製造指示一覧 生産ライン・工程

製造指示一覧

6 ステータス 製品 納期 検索

既存の絞り込みとページ移動

7 PO-2026-0001 ブレーキキャリパー

100 個 下書き 2026/10/05

生産ライン：L-001 組立ライン A

PO-2026-0002 生産ライン：未設定

1 PMAI メニュー

製造指示一覧

6 ステータス 製品 納期 検索

既存の絞り込みとページ移動

7 PO-2026-0001 ブレーキキャリパー

100 個 下書き 2026/10/05

生産ライン：L-001 組立ライン A

PO-2026-0002 生産ライン：未設定

Region	Screen / binding
1	Both: existing guarded header; dedicated Factory navigation to SCR-005
2	SCR-001: existing product/quantity/status inputs and unchanged field locks
3	SCR-001: lineId/current-selection summary; Draft Clear, non-Draft read-only
4	SCR-001: eligible q/page search/results, loading/empty/error; no historical options
5	SCR-001: line field error/summary; existing Save/Cancel and success notice
6	SCR-002: unchanged filters/URL/sort/paging; no new line filter or sort
7	SCR-002: non-sortable line column/card detail with current master or real null

Numbers are local to this addendum and do not renumber baseline items. At <640px form/picker controls stack and the list uses cards; >=640px uses the existing table. The line column has a plain scoped header, no sort button. Long names wrap; any desktop table horizontal scrolling stays inside the existing table container.

3. Field and assignment rules

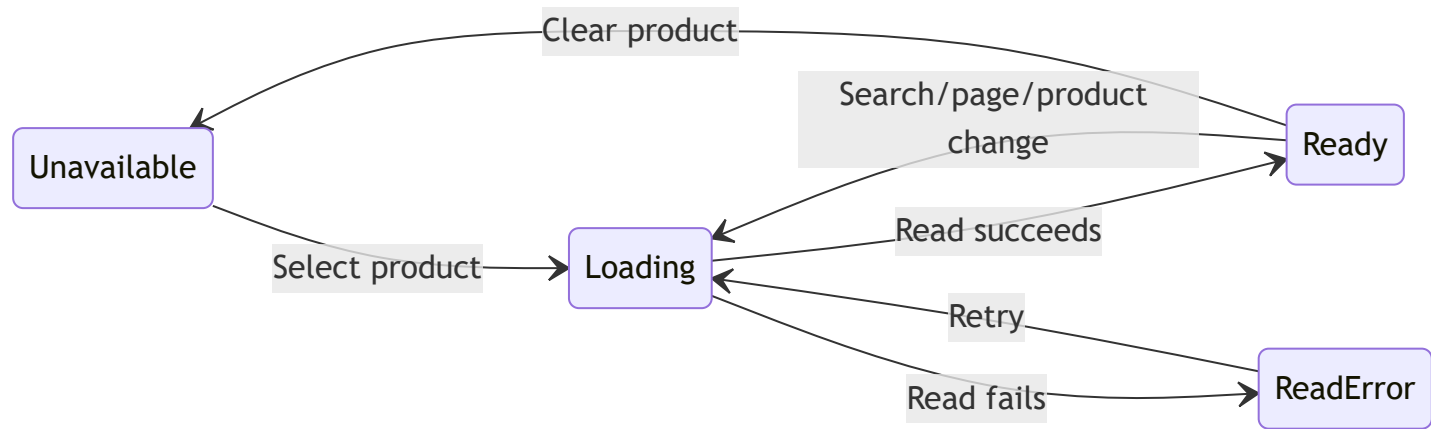
Existing order routes and the eligible-line API both require Admin or Operator through the existing ProductionOrderEditor policy. UI controls and direct routes enforce the same access; the server repeats authorization. No new role or broader read entitlement. An order sends only lineId for this addition, not line version, timing, unit confirmation or scheduling data.

Item	Local value / validation
lineId	UUID string or null; create default null. Include in dirty comparison. Lock based on origin saved status, not selected target status. Draft changing to InProgress may still select before Save.
Current line	Separate summary {id,code,name,isActive} from order response, or newly selected candidate. null displays unassigned. Retired master is visibly marked; saved selection need not appear on any candidate page.
Candidate search / paging	q trimmed <=100 code points; page 1–10000, fixed 50; Search/Clear reset page 1. Product change resets q/page and cancels obsolete reads. No picker state in order-list URL or persisted draft storage.
Timing hints	Show candidate minutesPerUnit as exact decimal text and visible product unit. It is a current configuration hint, not order history, reserved capacity, total or automatic duration. No new cross-unit sum.

Origin / requested action	Allowed result / server check
Create Draft, absent/null line	Save with no assignment. Non-null selection must be eligible for product.
Draft unrelated edit with unchanged product/line	Retain exact historical selection even if line/pair retired or confirmation stale; no new eligibility check unless starting.
Draft change line or product	New non-null exact pair must be eligible. Client always clears line on product change, including change back; no automatic substitute/restore. Explicit null clears with no line/pair check.
Draft → InProgress	Require non-null and recheck active product/line/pair plus unit revision under FN locks, including unchanged/omitted historical assignment.
Origin InProgress/Completed/Cancelled	lineId read-only; unchanged value/null or omission accepted during otherwise allowed edits. Changed assignment rejected LINE_LOCKED. Old non-Draft null stays null; no backfill requirement.

Order detail/list line fields cannot reveal pair retirement or confirmation revision. Do not label a saved line as fully eligible merely because isActive is true, and do not infer ineligibility from absence on a filtered/paged choice response. Starting always uses the server check.

4. Order picker state transitions



From	Event	To	Effect
Unavailable	Draft selects active product	Loading	Read API-PL-07; product change clears line first
Loading	Current read succeeds	Ready	Render page/empty; preserve selection separately
Loading	Current read fails	ReadError	No invented choices; show retry
ReadError	Explicit retry	Loading	New read; no order write
Ready	Apply search/page or active product change	Loading	Cancel old read; product change clears line
Ready	Clear product	Unavailable	Clear line and candidates; no request
Any picker state	Origin non-Draft, product cleared or inactive	Unavailable	Cancel read; retain original line on non-Draft; hide selectable panel
Ready	Select candidate / clear line	Ready	Local line draft only; dirty; never write order on selection

Six distinct-state chart edges match the first six rows; any-state and same-state cases are table-only. Unavailable means the selection panel is disabled/hidden, not that a historical line disappears. Picker state is orthogonal to the existing order form save/error state.

5. Client processing sequences

5.1 ProductionOrderPage / ProductionOrderForm / OrderLinePicker

Proposed metadata: Agent / created and modified 2026-10-01. Used services: existing getOrder/listProducts/createOrder/updateOrder, API-PL-07 eligible adapter, Japanese catalog and navigation guard. OrderLinePicker is a feature-local React composition, not a new framework.

Step	Processing / branches	Call / result
1	Load existing product/order baseline; decode required nullable line. Create initializes null. Existing 401/403/404 views remain; no editable fabricated baseline.	Existing page reads → initial form
2	Determine assignment lock from origin status. Non-Draft displays named line/null read-only with lock hint; Draft active product loads candidates. Retired product retains history but exposes no new line choices.	API-PL-07 or read-only view
3	Apply q/page with encoded productId, cancel superseded reads and guard response by product/request generation. A late result cannot restore a cleared/changed line. Search/page alone do not mark order dirty.	Eligible adapter → picker state
4	Select only a current-product candidate; show current selection independently of visible page. Keep baseline historical selection until explicit change; replacement/clear cannot reselect retired history as a new choice. Product change clears line.	Local lineId/summary → dirty draft
5	On Save preserve existing validation and status options. If requested Draft→InProgress has null, field error/focus, no mutation. Do not require candidate membership for unchanged historical Draft saves.	Form validation including lineId
6	New client sends explicit lineId UUID/null and existing order version/quantity fields. Keep exact order quantity number-token serializer; new line decimal-string contract must not replace it. Single save; no repeated pending mutation.	Existing POST/PUT order → committed response or error
7	Success keeps existing create→edit route and update-in-place behavior; response becomes authoritative baseline, including line. Notify with existing success catalog. No master mutation or second order save.	Existing onCreated / update baseline
8	On LINE_REQUIRED/INELIGIBLE/LOCKED retain draft and map errors.lineId to new field and summary. Offer explicit candidate refresh for Draft; refresh never clears/substitutes selected value. Retry requires explicit Save after correction.	Catalog + optional API-PL-07 read
9	Order-version conflict uses existing MSG-E009 with guarded reload; no merge/overwrite. Ambiguous write retains draft and disables another mutation until explicit order read reconciliation. Edit verifies id; lost create without known id cannot be auto-replayed or proven by similar fields; retain warning and use existing list for manual verification.	Existing order get/list; no automatic replay
10	Dirty guard includes line changes and covers Cancel/navbar/browser exits; pending save suppresses repeated submit/navigation. Existing discard dialog uses viewport centering and safe focus obligations. No localStorage/URL draft persistence.	Existing guard / discard dialog

5.2 ProductionOrderListPage / ProductionOrderTable

Proposed metadata: Agent / 2026-10-01. Used services: existing listOrders and formatters/catalog. Add the same nullable line projection to each row and mobile card; retain the original list state machine and API filters.

Step	Processing / branches	Result
1	Load using existing URL filters/sort/page/pageSize and cancellation; no lineId query parameter.	Unchanged loading/error/empty states
2	Render line code/name and retired marker only from response isActive. Real null displays unassigned; missing required property/integrity failure is not invented null.	Non-sortable line table cell / card detail
3	Keep order-number link/row navigation, quantity with unit, counts and keyboard sort unchanged. No per-row API-PL read, mixed-unit total or dashboard change.	Existing edit route / list controls

6. Error messages, accessibility and persistence boundary

New UI strings below are catalog proposals; existing headings/status/save/messages must match messages.ts. Only client text is translated; machine error codes and API types remain those of approved API/FN.

Binding	Japanese UI	English gloss
Field / null	生産ライン / 未設定	Production line / unassigned
Picker search / choose / clear	ラインコード・ライン名 / 選択 / 選択を解除	Line code/name / select / clear selection
Optional Draft help	下書きでは任意です。生産開始時に必要です。	Optional in Draft; required to start production
Locked help	下書き以外の製造指示では、生産ラインを変更できません。	Line cannot change outside Draft
History / retired	使用停止 / 現在の割当は履歴として保持されています。	Retired / current assignment retained as history
Empty eligible page	選択できる生産ラインはありません。	No selectable production lines
LINE_REQUIRED	生産を開始するには、生産ラインを選択してください。	Select a line to start production
LINE_INELIGIBLE	この製品を生産できる有効なラインを選択してください。	Select a valid line for this product
LINE_LOCKED	下書き以外の製造指示では、生産ラインを変更できません。	Cannot change a line outside Draft
Unknown write / verify	保存結果を確認できません。再送信せず、現在の状態を確認してください。 / 現在の状態を確認	Unknown save outcome; verify without replay / verify current state

400 line errors join existing field handling without replacing existing quantity/status/product codes; 409 remains the original numeric order-version conflict. Choice 404/error does not mark the order missing. 401 uses existing session expiry; 403 stops protected rendering/writes. Read retries are explicit. Server-confirmed LINE_BUSY rollback, when returned by FN integration, may be retried explicitly after Retry-After; generic timeout/500/lost or malformed write response is not rollback proof.

Accessibility: labeled line region with persistent optional/required/locked help; field error links and aria-describedby/invalid; candidate buttons named by code/name; polite loading/results and error alerts; scoped table header, descriptive mobile line text, non-color retired marker, visible focus and >=48px picker controls. Refresh/search focus result heading without disturbing a typed draft; selection returns to current-selection heading. Existing native discard dialog remains centered after scroll, keyboard contained, safe action first, Escape cancels and focus returns. Verify at 320px/200% zoom and keyboard later; static previews do not prove conformance.

Persistence remains the approved nullable production_orders.line_id and product/line composite reference. Detail/list project current master names with no N+1 reads; legacy null stays real null. Save/start eligibility uses the existing order transaction and product→line→pair SHARE locks through commit; original order xmin predicate prevents status-read races. No independent pair timing snapshot, unit conversion, capacity reservation or historical recomputation. Migration/write-pause/rollback limits belong to 005_DB; no SQL is executed here. Instrumentation adds the FN-owned ProductionLine.ValidateOrder child span inside existing order traces; eligible reads use ProductionLine.Eligible and approved bounded metrics. No raw ids/query/names/coefficients/versions/cookies/body diagnostics or new dependency.

7. Compatibility and verification viewpoints

Case	Expected result / later test
Old client missing lineId	Create Draft null; update preserves exact saved assignment. Draft start without valid non-null assignment fails, including existing Draft; coordinate rollout per DB.
Explicit null vs omission	Draft clear only for explicit null; non-Draft changed non-null→null fails; legacy null→null allowed.
Retired/stale pair and historical line	Unchanged history saves; new/changed selection and start reject; line isActive alone is insufficient.
Product change and late choice response	Clear line immediately; no stale response restore/substitute; server rejects incompatible retained UUID.
Unit ABA / retirement / order-version race	Server locks/checks and final xmin make save/start atomic; retain draft on error.
Line list presentation	Current names/retired/null displayed identically in table/cards; unchanged list URL/sort/counts/quantity units.
Read/error/unknown/accessibility	No fabricated choice/line; no automatic replay; 401/403/404 scopes correct; field focus/keyboard/mobile checked later.

Open business decisions: none. These are design expectations, not executed application tests. Actual document/visual verification is recorded in [evidence](#). Submit this one design package for explicit review. After approval, revision 1 step 11 reconciles the full family/security and prepares the test plan; step 12 presents implementation revision 2 for separate approval. No application code, migration, push, PR, merge or deployment is authorized by this document.